

Beyond Action Recognition: Extracting Meaningful Information from Procedure Recordings

Tim J. Schoonbeek*
Eindhoven University of Technology

Hans Onvlee
ASML Research

Pierluigi Frisco
ASML Research

Peter H.N. de With
Eindhoven University of Technology

Fons van der Sommen
Eindhoven University of Technology

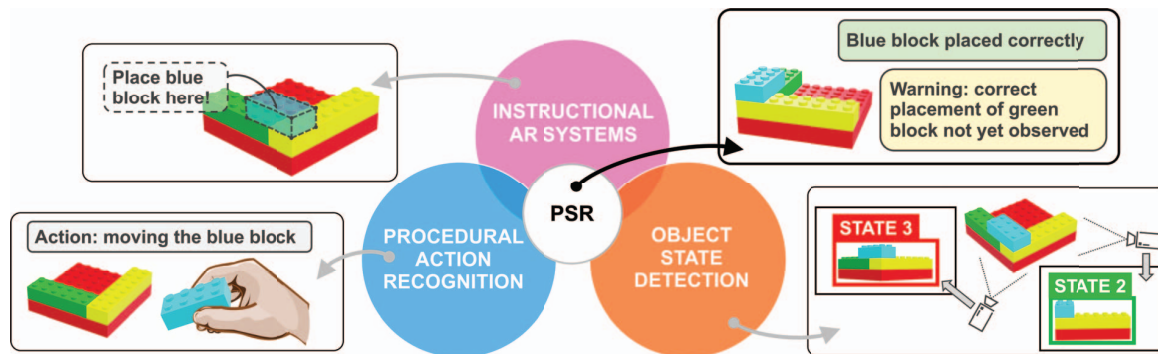


Figure 1: Procedure step recognition lies at the intersection of three research fields. It aims at recognizing the completion of procedural actions and determines whether those actions, as well as their order, are in line with the expected procedure.

ABSTRACT

Understanding procedural actions is important, as it can be used to automatically analyze the execution of a procedure and provide assistance to users by warning for potential mistakes or forgotten steps. However, current approaches require a rigid, step-by-step execution order, laborious and impractical datasets. Furthermore, they are unreliable to variations in viewpoint, or measure the performance of actions rather than the actual completion of actions. To address these limitations and stimulate research in this field, this work proposes the novel task of *procedure state recognition (PSR)* together with a set of evaluation metrics.

1 INTRODUCTION

Automated understanding of procedural actions, such as performing maintenance on a motor, is worthwhile for various reasons and has promising applications in augmented reality (AR). It can be used to warn the user of (potential) mistakes and forgotten steps, communicate relevant instructions to the person executing the procedure, or automatically generate procedural logs [6]. Understanding procedural actions is a difficult task in computer vision. Firstly, often limited visual difference between subsequent steps is present. Additionally, a high degree of symmetry and a low degree of textual variation further complicates this issue. Furthermore, it is typically not feasible to collect a vast amount of video data for each specific procedure, as they are often infrequent and specialized. Finally, procedures can generally be correctly completed via many possible execution orders. Therefore, it is insufficient to look whether a single, pre-defined step is completed correctly.

Existing approaches to procedural action understanding are outlined in Fig. 1, and can be divided into three sub-groups.

Instructional AR systems: Several instructional AR implementations exist, that aim at teaching a task to an operator [2, 9, 10]. These approaches do not allow flexible execution order of the procedure, even in places where such flexibility is acceptable and indeed encouraged. This rigidity is especially a notable limitation when the

system is used by experienced operators, as they likely have their own preferred (and potentially more efficient) procedure flow.

Procedural action recognition (PAR): In PAR, the objective is to classify which action is being performed in a video segment [5, 7]. Similarly, temporal action segmentation (TAS) also aims to recognize actions [3], but takes the entire procedure video (instead of shorter clips) as input and predicts frame-wise action class labels. PAR and TAS both allow for a flexible execution order, but aim at recognizing the execution of actions, rather than the completion of those actions. The correct completion, or measurable success, of procedural actions is significantly more important than the mere performing of actions, especially in industrial settings.

Object state detection (OSD): OSD refers to the recognition and localization of a specific object and its state within an image. Recent works have demonstrated promising advances in this field, even for fine-grained state recognition [4, 8]. However, OSD has two significant limitations. Firstly, OSD approaches do not explicitly leverage procedural information, making it unreliable for identifying objects that appear similar in different states. Secondly, the computer vision models require large, laborious datasets, and the dataset size grows exponentially with procedure complexity. It remains unclear whether such approaches still perform well for more complex procedures. Furthermore, any modifications to the procedure necessitate updating the entire dataset and retraining the OSD approach, which is a costly and time-consuming process.

To address the limitations of the aforementioned tasks, we propose the novel task of *procedure state recognition (PSR)*. This task aims at recognizing the completion of actions (rather than merely their execution), explicitly utilizing procedural information and accommodating for flexibility in the execution order, when the procedure allows it. Furthermore, we provide relevant evaluation metrics for quantifying the performance of a system towards this task. Finally, we demonstrate a PSR implementation on a simplified assembly task in the supplementary material.

2 PROCEDURE STATE RECOGNITION

By proposing the PSR task, we hope to encourage researchers to develop methods to automatically recognize the completion of steps,

*e-mail: t.j.schoonbeek@tue.nl

rather than the (partial) execution of activities, and to explicitly leverage procedure knowledge.

Task definition. In PSR, the objective is to extract an estimation of the procedure steps correctly performed by a person, based on sensory inputs X and a descriptive set of the procedural actions to be performed P . We emphasize recognizing the followed procedure, rather than a successful end result. Thus, the goal of PSR is to recognize the correctly performed steps and their execution order, which can subsequently be used to determine the correctness of the procedure execution. The predicted procedure execution $\hat{y}$, given the inputs and some PSR model $\mathcal{F}$, is defined such that

$$\hat{y} = \mathcal{F}(X, P). \quad (1)$$

Here, “predicted” does not refer to the prediction of a future step, based on observed sensory data, but rather the prediction of a procedure step completion by a model based on the given inputs. Each unique action $a_i \in P$ contains information describing this specific action, and can be different for varying approaches to PSR. For instance, a_i can contain a description, a template image of what the completed step is supposed to look like, or relative positions between two components. Sensory inputs X may comprise of camera images, depth maps, or even user interaction with the system, during the execution of a procedure.

The predicted order in which someone has performed the procedure steps is an ordered list of pairwise distinct elements, $\hat{y}$, where each element is the predicted completion of an action $a_i \in P$, at time $\hat{t}_{(i)}$, and with a confidence $c_{(i)}$. The first predicted step is not necessarily the completion of action $a_0 \in P$, but rather the first step that a person actually has completed. Moreover, no action a_i can be predicted to be completed more than once. Finally, not every action a_i has to be predicted. The order in which the steps are actually completed, is the ground-truth execution order y . Note that the length of y and $\hat{y}$ are not necessarily equal, as not all executed steps may be recognized. Moreover, $\hat{y}$ can have false positive predictions.

Evaluation metrics. To quantify the performance of a PSR system and compare one system to another, evaluation metrics are required. We propose to use three metrics.

Procedure order similarity (POS): The POS is a variant of the Levenshtein (Lev) edit distance, commonly used in TAS [3]. Two changes to this existing metric are proposed, namely to weigh substitutions with a factor of 2, rather than considering them equal to insertions and deletions. This essentially eliminates substitutions, as a deletion followed by an insertion is of equal edit distance. The exclusion of substitutions is proposed to prevent the metric from favouring models with many false positives. The second proposed modification is to use the Damerau-Levenshtein (DamLev) [1] edit distance, as DamLev counts the transposition of two adjacent predictions as only one edit (as opposed to two edits for Lev), which is intuitive for determining the similarity when order matters. With these modifications, we define the POS edit distance as

$$\text{POS} = 1 - \min\left(\frac{\text{DamLev}(y, \hat{y})}{|y|}, 1\right), \quad (2)$$

where $\text{DamLev}()$ is the weighted DamLev edit distance function.

F_1 score: The F_1 score is a common metric in object state detection and action recognition tasks. It defines a joint score based on false positive (FP), false negative (FN), and true positive (TP) predictions. For the PSR task, a FP is defined as a procedure step completion that is predicted prior to the ground truth completion of action a_i , or if a_i is not at all completed. An FN is defined as a step which has indeed been completed, but is not represented in $\hat{y}$. Finally, a TP is defined as the prediction of a procedure step completion, recognized at or after the actual completion of a_i .

Average delay (τ): The average delay quantifies the time between the ground-truth completion of a step, and the corresponding true

positive recognition of that step. Since FNs and FPs have either undefined or negative delays, they are excluded from τ .

Individually, each metric provides limited information regarding the overall performance of a system. For instance, a system that predicts nothing until the last frame and then predicts all steps in P in a random order, has a perfect F_1 score (if all steps in P occur in the ground truth). However, the POS and τ metrics will indicate poor performance, which highlights the need to use a combination of multiple metrics. Nonetheless, for specific applications, one of the metrics may be deemed irrelevant and can be omitted accordingly.

3 DISCUSSION AND FUTURE WORK

An aspect that is currently not defined in PSR, is explicitly recognizing erroneous actions, or action executions. The proposed task definition of PSR only takes information regarding the proper execution of tasks as input (P). It would be an interesting research challenge to also take into account some input Q , representing a set of states or actions which are explicitly not expected. Secondly, the proposed evaluation metrics do not explicitly handle the simultaneous completion of tasks. We have proposed to weigh substitutions by 2, rather than weighing all edits equally as originally proposed [3], motivated by the fact that with equally weighted substitutions, a model that predicts a random, erroneous step would score higher than one that does not predict any step. However, the weights can be chosen empirically or dependent on a different motivation than ours, indicating that the selection of the weights may be sub-optimal. Nonetheless, it is important to have a standardized set of evaluation metrics that can be compared across applications and implementations.

In future work, we plan to demonstrate the applicability of procedure state recognition on a complex system, and release a challenging public dataset for PSR in an industrial setting. With this and the work presented here, we aim to stimulate further research in the field of procedural action understanding, paving the way to advanced support during the execution of procedures.

REFERENCES

- [1] F. J. Damerau. A technique for computer detection and correction of spelling errors. *Communications of the ACM*, 7(3):171–176, Mar. 1964.
- [2] J. D. Jones, C. Cortesa, A. Shelton, B. Landau, S. Khudanpur, and G. D. Hager. Fine-grained activity recognition for assembly videos. *IEEE Robotics and Automation Letters*, 6(2):3728–3735, Apr. 2021.
- [3] C. Lea, A. Reiter, R. Vidal, and G. D. Hager. Segmental spatiotemporal cnns for fine-grained action segmentation. In *European conference on computer vision*, pages 36–52. Springer, 2016.
- [4] H. Liu, Y. Su, J. Rambach, A. Pagani, and D. Stricker. TGA: Two-level group attention for assembly state detection. In *Proc. ISMAR-Adjunct*, pages 258–263, Los Alamitos, 2020. IEEE.
- [5] F. Ragusa, A. Furnari, S. Livatino, and G. M. Farinella. The MEC-CANO dataset: Understanding human-object interactions from ego-centric videos in an industrial-like domain. In *Proc. WACV*, pages 1569–1578, Los Alamitos, 2021. IEEE.
- [6] T. J. Schoonbeek, P. Frisco, H. Onvlee, F. van der Sommen, et al. Augmented reality for automatically generating robust manufacturing and maintenance logs. In *Proc. LIM*, Springfield, 2022. IS&T.
- [7] F. Sener, D. Chatterjee, D. Shelepov, K. He, D. Singhania, R. Wang, and A. Yao. Assembly101: A large-scale multi-view video dataset for understanding procedural activities. In *Proc. CVPR*, pages 21096–21106, Los Alamitos, 2022. IEEE.
- [8] Y. Su, J. Rambach, N. Minaskan, P. Lesur, A. Pagani, and D. Stricker. Deep multi-state object pose estimation for augmented reality assembly. In *Proc. ISMAR-Adjunct*, pages 222–227, Los Alamitos, 2019. IEEE.
- [9] G. Westerfield, A. Mitrovic, and M. Billinghurst. Intelligent augmented reality training for motherboard assembly. *International Journal of Artificial Intelligence in Education*, 25(1):157–172, Mar. 2015.
- [10] X. Yin, X. Fan, W. Zhu, and R. Liu. Synchronous AR assembly assistance and monitoring system based on ego-centric vision. *Assembly Automation*, 39(1):1–16, Apr. 2018.